D2 - SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

AgriUAV Platform: Precision Drone Dispatching Platform in the Mekong Delta

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Group	Group 3
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1. Introduction

1.1 Purpose

This document is the Software Requirements Specification (SRS) for the AgriUAV Platform - Precision Agricultural Drone Spraying Coordination Platform for the Mekong Delta, developed in accordance with IEEE Std 830-1998.

Purpose of this SRS:

- Provide complete and clear requirements specifications for the development team (System Designer, System Analyst) to design and implement the system according to business needs.
- Serve as a baseline working agreement between the analysis team and stakeholders (Cooperatives, Plant Protection Department) on the scope and features of the system; final stakeholder sign-off is pending confirmation in a future pilot.
- Serve as the basis for building test cases, traceability matrix, and subsequent deliverables (D3 Models, D4 SDD).

1.2 Scope

System Name: AgriUAV Platform

AgriUAV Platform is a Web (Dashboard) and Mobile App platform that helps Agricultural Cooperatives in the Mekong Delta support the scheduling, coordination, and near real-time monitoring of plant-protection drone fleets with precision.

In-Scope: Manage a drone fleet per cooperative against a reference workload of up to 50 concurrently monitored drone telemetry streams (configurable, not a hard-coded limit); spray scheduling; field mapping; rule-based assignment recommendation with HTX Manager confirmation; near real-time tracking via operator mobile GPS or structured simulated telemetry; geofence proximity/crossing alerts (including a 50 m buffer zone around registered high-voltage power lines - see FR12); compliance-support report export for Decree 288/2025/ND-CP.

Out-of-Scope: Direct drone control (fly-by-wire); AI pest detection; integration with irrigation systems or other agricultural machinery; integration with actual drone hardware; financial/invoicing management; UTM flight permit applications.

1.3 Definitions, Acronyms, Abbreviations

See Appendix A - Glossary at the end of this document.

1.4 References

- Nguyen, D.L. et al. (2025). 'The Effect of Farm Size on the Decision to Adopt Digital Technology: UAVs in Rice Production in Vietnam.' Research on World Agricultural Economy.
- Decree 288/2025/ND-CP - Regulation of unmanned aerial vehicle flight operations (effective 05/11/2025).
- Vietnam News (2024). 'Use of agricultural drones on the rise in the Mekong Delta.'
- IEEE Std 830-1998 - IEEE Recommended Practice for Software Requirements Specifications.
- D1 AgriUAV Platform - Project Proposal v1.5 (ITA301 Summer 2026, Group 3).
- Tuoi Tre Newspaper. (2024). Agricultural drone incident involving 110kV transmission line contact, causing outages for 76,000 customers in Long An Province. [https://tuoitre.vn/drone-phun-thuoc-tru-sau-va-vao-day-110kv-gay-mat-dien-o-5-huyen-tai-long-an-20241014173928997.htm]

1.5 Overview

This SRS document is organized according to IEEE 830-1998 and includes the following sections:

- (2) Overall system description
- (3) Functional (FR) and Non-Functional (NFR) requirements specification
- (4) External interfaces
- (5) Use Cases
- (6) User Stories
- Glossary Appendix.

2. Overall Description

2.1 Product Perspective

AgriUAV Platform is designed to address the gap at the **Platform Layer** in the agricultural UAV ecosystem in the Mekong Delta:

Layer	Current State	AgriUAV Role
Hardware	DJI, XAG, YANMAR - saturated market	Current: drone position is received indirectly via the pilot's smartphone GPS (see Assumption in Section 2.5 and Section 4.2). Future: direct MAVLink telemetry integration with DJI/XAG drones.
Service	Cooperatives (HTX), private entities - manual operation (Zalo/Excel)	Digitize and automate operations
Platform	**AgriUAV Platform - Platform-Layer gap identified from reviewed evidence; addressed by this design**	

2.2 Product Functions (Summary)

- Field Mapping: Digitize field maps with GPS; support formula-based pesticide dosage calculations that require authorized agronomist or HTX Manager confirmation before dispatch.
- Scheduling: Book spraying sessions online, automatic schedule-conflict checking and order acknowledgement; the HTX Manager confirms assignment where required (consistent with UC01 and UC02).
- Fleet Dispatch: Recommend drone–pilot assignments using rule-based checks (drone status, battery, location, maintenance, and pilot certification validity); the HTX Manager reviews and confirms the final assignment.
- Near Real-time Monitoring: Track drone positions (via operator mobile GPS or simulated telemetry), geofence alerts, monitor battery and pesticide levels.
- Compliance Support: Store flight logs and export Plant Protection Department reports per Decree 288/2025 (exact official format subject to authority confirmation).
- Notification: Send SMS/Zalo status updates to farmers.

2.3 User Characteristics

User	IT Skill	Characteristics & UX Requirements
Farmer	Low (basic smartphone)	Average age ~51 (Nguyen et al., 2025). The app needs intuitive icons, large text ≥ 16pt, ≤ 5 step booking, offline mode.
HTX Manager	Medium (Excel, Zalo)	Needs an overview Dashboard, calendar view, 1 click reports. Uses Web on desktop.
Operator	Medium (smartphone)	Uses Mobile App in the field. Needs GPS check-in, clear alerts, offline operation.
Agronomist	High (Excel, GIS)	Needs to configure spray formulas, detailed field mapping, view historical data.
Admin	High (system administration)	System administrator: manages user accounts and role assignments, system configuration, and audit-log

User	IT Skill	Characteristics & UX Requirements
		oversight. Uses the Web Dashboard on desktop; full access per RBAC (FR18).

2.4 Constraints

- The system only assigns pilots holding valid flight certificates under Decree 288/2025/ND-CP.
- The Mobile App must support basic offline operation (view schedule, check-in/out) when 4G is unavailable in the field.
- Flight log data must be retained for at least 5 years (working assumption pending confirmation of the specific article reference in Decree 288/2025/ND-CP or related regulations).
- The system does not directly interfere with drone hardware (no fly-by-wire, no autopilot).
- Reference workload: the design is validated against one cooperative managing up to 50 concurrently monitored drone telemetry streams; the registered fleet size is configurable per cooperative and is not a hard-coded limit. Nationwide scale is outside the initial evaluation scope.

2.5 Assumptions and Dependencies

- Assumption: Drones have been registered and assigned valid identifiers under Decree 288/2025 before being added to the system.
- Assumption: Pilots already hold flight certificates issued by competent authorities before an Operator account is created.
- Assumption: MAVLink telemetry integration with real drones is OUT OF SCOPE for the initial phase; drone position is updated via the pilot's Mobile App GPS or structured simulated telemetry.
- Dependency: The system depends on a third-party Weather API (OpenWeatherMap). If the API is down, weather alert features will not function.
- Dependency: The Zalo notification feature depends on the Zalo Official Account API.

3. Specific Requirements

3.1 Functional Requirements (FR)

The table below lists 21 Functional Requirements classified by module, with Priority following MoSCoW (Must / Should / Could). Source specifies the originating stakeholder or pain point of the requirement.

ID	Requirement Description	Module	Priority	Source
FR01	The system allows the HTX Manager to create, edit, and store field maps with GPS coordinates, area (ha), and crop type.	Field Mapping	Must	HTX Manager, Pain Point Step 1
FR02	The system calculates and displays pesticide dosages for each field zone based on area, pest type, and spray formulas configured by the Agronomist. The final formula and dosage require confirmation by an authorized Agronomist or HTX Manager before mission dispatch.	Field Mapping	Must	Agronomist, Plant Protection Regulations
FR03	The system provides a spraying booking function for farmers: choose field, spray date/time, pesticide type, and confirm the order in ≤ 5 steps.	Scheduling	Must	Farmer, Pain Point Step 2
FR04	The system automatically checks for schedule conflicts and displays warnings when multiple spraying orders overlap in time on the same area.	Scheduling	Must	HTX Manager, Pain Point Step 2
FR05	The HTX Manager can view, edit, and confirm spraying schedules in a calendar view (by day / week / month).	Scheduling	Should	HTX Manager
FR06	The system automatically suggests drone-pilot assignments for each spraying order based on: drone operational status (Available / In-Flight / Under Maintenance), remaining tank capacity, battery level, proximity to the field, pilot certificate validity, and pilot schedule availability; the HTX Manager reviews and confirms the final assignment. (Rule set is the source of truth for UC02.)	Fleet Dispatch	Must	HTX Manager, Pain Point Step 3
FR07	The system manages pilot records including full name, flight certificate number, certificate expiration date, flight history, and operational status.	Fleet Dispatch	Must	Decree 288/2025/ND-CP - FR-REG01
FR08	The Operator/Pilot receives the flight assignment notification via the Mobile App with complete information: field, GPS coordinates, pesticide dosage, and required time.	Fleet Dispatch	Must	Operator/Pilot

ID	Requirement Description	Module	Priority	Source
FR09	The Pilot can perform on-site check-in (GPS-verified) and check-out upon completion; the system automatically records the actual time and location. Check-in is only accepted within +30 or -30 minutes of the scheduled spraying time (business rule enforced at UC03).	Fleet Dispatch	**Must**	HTX Manager, operator-presence verification
FR10	The system displays the position and status of all active drones on a near real-time map, updated at least every 3 seconds (target refresh interval). Within the MVP scope, position is derived from the operator's mobile-app GPS (treated as a mission-level proxy for drone position) or from structured simulated telemetry; direct MAVLink integration with drone hardware is post-MVP (see Section 2.5 Assumption).	Monitoring	**Must**	HTX Manager, Pain Point Step 4
FR11	The system automatically issues alerts (push notification + dashboard display) when it detects: (a) drone battery below 20%, (b) pesticide below the completion threshold, (c) drone approaching or crossing the geofence boundary, (d) adverse weather - wind speed ≥ 3 m/s (an assumed threshold, pending expert confirmation) or rain forecast within the next 30 minutes (see UC06). Within MVP scope, battery and pesticide levels are obtained from operator manual entry on the Mobile App or from structured simulated telemetry; native sensor telemetry via MAVLink is post-MVP (see Section 2.5).	Monitoring	**Must**	Safety, Pain Point Step 4
FR12	The system configures geofence boundaries based on field boundaries (including a 50 m buffer zone around marked high-voltage power lines) and issues information-level alerts when a drone approaches or crosses a boundary. The system does not directly control the drone, firmware, or autopilot; the pilot remains responsible for flight safety.	Monitoring	**Must**	Safety, Long An 110kV incident (Tuoi Tre, 2024)
FR13	The system automatically stores complete flight logs including drone identifier, pilot ID, flight time, GPS coordinates, and volume of pesticide sprayed.	Compliance	**Must**	Decree 288/2025/ND-CP - FR-REG01
FR14	The system exports seasonal reports (PDF/Excel) with complete information: spray area, pesticide type, dosage, and confirmation that the pilot holds a valid certificate. The layout is aligned with the Plant Protection Department's reporting fields; the exact official format	Compliance	**Must**	Plant Protection Department - FR-REG02, Decree 288

ID	Requirement Description	Module	Priority	Source
	requires confirmation by the competent authority.			
FR15	The HTX Manager can retrieve the spray history of each field by time range and export data in a VietGAP/GlobalGAP-aligned format (project-defined minimum field set, pending stakeholder validation with export companies) for sharing with contracted export partners.	Compliance	**Should**	Agricultural export companies
FR16	The system manages pesticide inventory: stock entry, stock-level tracking, low-stock alerts (below configured threshold), and verification that pesticides are on the Plant Protection Department's approved list.	Inventory	**Should**	HTX Manager, Agronomist
FR17	The system sends spraying-order status updates to farmers via SMS and/or Zalo: booking confirmation, spraying started, spraying completed.	Notification	**Should**	Farmer
FR18	The system enforces role-based access control with 5 roles: Admin (full access), HTX Manager (manages their own cooperative), Operator (views personal schedule + check-in/out), Farmer (books spraying + views progress), Agronomist (configures spray formulas and dosage rules; confirms dosages).	User Management	**Must**	All stakeholders
FR19	The Mobile App and Web Dashboard support dark mode, which can be toggled manually or synced with the device's system theme.	UX	**Could**	Operator (early-morning / late-evening flights, anti-glare)
FR20	The system supports multiple languages at minimum Vietnamese (default) and English with runtime switching that does not require re-login.	i18n	**Could**	Export companies, foreign investors
FR21	The system allows the HTX Manager and Admin to register and manage drone master records: drone identification code (as assigned under Decree 288/2025/ND-CP), model specification, tank capacity (litres), and operational status (Available / In-Flight / Under Maintenance). The status field is updated by the system as missions are assigned and completed. These records are the authoritative data source for FR06 assignment logic and FR13 flight-log generation.	Fleet Management	**Must**	HTX Manager, D1 §1.3 Function #1, Decree 288/2025/ND-CP

Must	Mandatory - the system is incomplete without it	**Should**	Important - should be in the MVP	**Could**	Optional - only if time allows

3.2 Non-Functional Requirements (NFR)

The table below specifies **7 Non-Functional Requirements** with specific quantitative thresholds, calibrated to the actual context of the Mekong Delta.

ID	Category	Criterion	Quantitative Threshold	Rationale / Source
NFR01	**Performance**	Drone position update on the map	Update cycle ≤ 3 seconds for up to 50 concurrent drones. Data transmission latency <500ms under normal 4G conditions.	DJI Agras T40 drone speed: ~5-7 m/s; 3 seconds is sufficient for safety alerts
NFR02	**Reliability**	System availability during peak season	Uptime ≥ 99% (allowing max downtime ~7.2 hours/year). RTO (Recovery Time Objective) ≤ 15 minutes after an incident.	Peak rice season: 30-60 days, extended downtime is unacceptable
NFR03	**Usability**	Farmer completes a spraying booking	First-time users complete the booking task in ≤ 5 minutes with no assistance. The Mobile App interface supports font ≥ 16pt and intuitive icons for users with limited IT experience (average age ~51).	Nguyen et al. 2025: Mekong Delta farmers' average age is 51, with limited IT experience
NFR04	**Scalability**	Drone fleet size and concurrent users	MVP acceptance baseline: supports up to 50 concurrent drones for one cooperative; transactional throughput target ~20 requests/second steady-state. Flight log data retention: a working assumption of at least 5 years pending confirmation of a specific article reference in Decree 288/2025/ND-CP or related regulations. Future-cluster planning assumption (NOT an MVP acceptance threshold): a representative deployment cluster of 10 cooperatives concurrently, giving ~167 req/s for position updates plus headroom for booking/report traffic (target ~200 req/s); this is an illustrative basis for capacity planning consistent with the SAM of ~1,215 drones across 3 provinces (Section 2.2 / D1).	Reference workload: up to 50 concurrently monitored drones per cooperative (configurable, not a hard limit). Sizing assumption: a representative deployment cluster serves 10 cooperatives concurrently (illustrative basis for capacity planning, not a hard scope ceiling -actual SAM covers ~1,215 drones across 3 provinces per Section 2.2 D1). Calculation: 50 drones × 10 cooperatives × 1 position update per 3s ~ 167 req/s → 200 req/s provides ~20% headroom for

ID	Category	Criterion	Quantitative Threshold	Rationale / Source
				booking/report traffic.
NFR05	**Security**	Data security and access control	Security outcomes (solution-neutral): data in transit must be encrypted using an industry-standard transport protocol (current minimum baseline: TLS 1.3); sensitive data at rest must be encrypted with a strong symmetric cipher (current minimum baseline: AES-256); authentication must use stateless token-based mechanisms with refresh capability (current minimum baseline: JWT + Refresh Token); access control follows RBAC with 5 roles per FR18; all security-relevant events are written to an immutable audit log. Specific technology choices may be finalized in D4 Design.	Decree 288/2025 on airspace data security; OWASP Top 10
NFR06	**Availability**	Basic offline operation in the field when 4G is unavailable	The Mobile App maintains offline mode: view cached flight schedule, offline check-in/out (synced when connection is restored). Automatic sync within ≤ 30 seconds when connectivity returns.	Mekong Delta reality: remote areas often lack 4G; pilots need to check in directly in the field
NFR07	**Maintainability**	Maintainability and system update capability	Hotfix deployment time ≤ 4 hours. Loosely coupled modules so that updating one module does not affect others. Test code coverage ≥ 70%.	Decree 288 may change -> the Compliance module needs to be updated quickly

4. External Interface Requirements

4.1 User Interfaces

- Web Dashboard (HTX Manager, Admin): React.js / Vue.js, supports the latest Chrome, Firefox, and Safari browsers. Responsive from 1280px. Map integration via Leaflet.js.
- Mobile App (Farmer, Operator): React Native / Flutter, supports Android ≥ 8.0 and iOS ≥ 14.0 Offline first architecture.
- Accessibility requirements: Font ≥ 16pt on mobile, contrast ratio ≥ 4.5:1, icons with label text.

4.2 Hardware Interfaces

- Smartphone GPS module: accuracy ≤ 5m for check-in verification.
- Smartphone camera: capture on site photos attached to completion reports (optional).
- MAVLink telemetry interface: out of MVP scope recorded in assumptions for a future version.

4.3 Software Interfaces

Interface	Description & Technical Specification
GPS / Location Service	The Mobile App uses native GPS (iOS CoreLocation / Android LocationManager). Required accuracy: ≤ 5m. Used to verify pilot check-ins and display drone positions.
Weather API	Weather REST API integration (OpenWeatherMap or equivalent). Poll every 15 minutes. Response: wind speed, rainfall, 2 hour forecast. Format: JSON.
SMS / Zalo Notification	Send SMS via Twilio or VNPT SMS Gateway. Send Zalo OA messages via the Zalo Official Account API. Triggers: successful booking, spraying started, spraying completed.
MAVLink Telemetry (future)	MAVLink v2 protocol to receive telemetry from DJI/XAG drones if hardware is integrated (outside current D1 scope - recorded in assumptions). Data: GPS, altitude, battery, speed.
Map / GIS	Leaflet.js or Google Maps SDK to display field maps and drones. Supports drawing polygons (field boundaries) and displaying drone GPS trails.
Export Format	Exported reports: PDF (pdfmake/iTextSharp) and Excel (Apache POI/ExcelJS). Template aligned with the Plant Protection Department's reporting fields (exact official format subject to authority confirmation). VietGAP/GlobalGAP: a project-defined JSON/CSV interchange format with a minimum field set, pending stakeholder validation with contracted export companies.

4.4 Communication Interfaces

- Backend and Frontend: REST API over HTTPS (TLS 1.3). JSON format.
- Near real-time updates: WebSocket (Socket.IO) for drone position tracking (≤ 3s target update cycle).
- Push notifications: Firebase Cloud Messaging (FCM) for Android/iOS.
- SMS: HTTPS POST to Twilio API / VNPT SMS Gateway.

5. Use Cases

This section describes 10 main Use Cases of the AgriUAV Platform, including the main flow and ≥ 2 alternative flows for each critical UC.

UC01 - Book a Spraying Session

Use Case ID	UC01
Use Case Name	Book a Spraying Session
Primary Actor	Farmer
Secondary Actor	HTX Manager (confirmation)
Preconditions	1. Farmer is logged in. 2. The Farmer’s field has been created in the system. 3. At least 1 drone in the fleet is available.
Postconditions (Success)	The spraying order is saved with status 'Pending confirmation'. The HTX Manager receives a notification to assign a drone.
Postconditions (Failure)	If the Farmer does not re-confirm the order within 24 hours of creation, the system automatically cancels the order, changes its status to 'Cancelled - Confirmation timeout', notifies the Farmer, and releases the reserved time slot. If the system cannot create the order due to a technical fault (DB error, pesticide-catalog API unavailable), the status remains unchanged and the Farmer receives an error notification.
Main Flow	1. The Farmer selects a field from the list. 2. The Farmer selects the desired spray date and time. 3. The system checks availability and displays free time slots. 4. The Farmer selects a pesticide from the approved list. 5. The Farmer confirms the booking; the system creates the order and sends SMS/Zalo notifications.
Alternative Flow	Alt 3a - Schedule conflict: The system warns about the conflict and suggests the 3 nearest available time slots. Alt 4a - Pesticide not on the approved list: The system displays a red warning and does not allow order confirmation.
Business Rule	Bookings must be made at least 24 hours before the spraying time. If forecast wind speed ≥ 3 m/s at the requested time slot, the system displays a warning and blocks Farmer self-confirmation; only the HTX Manager may override with a risk acknowledgement recorded in the audit trail (soft constraint - consistent with UC06).
Priority	Must - Linked: FR03, FR04, FR17

UC02 - Drone–Pilot Assignment Recommendation for a Spraying Order

Use Case ID	UC02
Use Case Name	Drone–Pilot Assignment Recommendation for a Spraying Order
Primary Actor	HTX Manager
Secondary Actor	System (assignment suggestion)
Preconditions	1. A spraying order has been created (UC01) and is in 'Pending confirmation' status. 2. At least 1 drone and 1 valid pilot (with a non-expired certificate) are available.

Postconditions (Success)	A drone and a pilot are assigned. The Pilot receives a notification on the Mobile App. The order moves to 'Assigned'.
Main Flow	1. The HTX Manager opens the list of spraying orders pending assignment. 2. The HTX Manager selects an order to process. 3. The system automatically suggests the most suitable drone (based on: nearest location, sufficient battery, remaining tank). 4. The system automatically suggests a suitable pilot (valid certificate, free schedule). 5. The HTX Manager confirms or modifies the suggestion. 6. The system updates the order status to 'Assigned' and notifies the Pilot.
Alternative Flow	Alt 3a - No drone available: The system notifies the user and suggests rescheduling to a different time slot. Alt 4a - Pilot lacks a valid certificate: The system hides that pilot's name and displays the reason.
Business Rule	Pilots must hold a valid flight certificate under Decree 288/2025. Drones under maintenance are not assigned.
Priority	Must - Linked: FR06, FR07, FR08, FR13

UC03 - On-site Check-in (GPS-verified)

Use Case ID	UC03
Use Case Name	On-site Check-in (GPS-verified)
Primary Actor	Operator / Pilot
Secondary Actor	---
Preconditions	1. The Pilot has received the flight assignment (UC02) and is logged into the Mobile App. 2. The Pilot is on-site at the field (within 100m of the field's GPS coordinates). 3. The drone has been prepared and is ready to fly.
Postconditions (Success)	Check-in is recorded with timestamp and GPS. The HTX Manager sees the near real-time update on the dashboard. The geofence monitoring is activated.
Main Flow	1. The Pilot opens the Mobile App and selects today's spraying order. 2. The Pilot taps the 'On-site Check-in' button. 3. The system verifies the Pilot's GPS position (within 100m of the field's coordinates). 4. The Pilot enters pre-flight parameters: pesticide level in the tank, drone condition. 5. The system records the check-in time and GPS position and updates the order status to 'In progress'.
Alternative Flow	Alt 3a - Pilot’s GPS is outside the 100m radius: The system blocks Pilot self-confirmation, sends an override-approval request to the HTX Manager, and requires the Pilot to submit a stated reason (and an optional evidence photo). Only the HTX Manager may approve the override; the approval, reason, and timestamp are recorded in the audit trail (consistent with FR18 RBAC). Alt 4a - Offline (no 4G): The system stores the check-in locally and syncs once connectivity returns (≤ 30 seconds).
Business Rule	GPS verification radius: 100m. Check-in is only allowed within +30 or -30 minutes of the scheduled spraying time.
Priority	Must - Linked: FR09, FR10, FR12

UC04 - Near Real-time Spraying Progress Monitoring

Use Case ID	UC04
Use Case Name	Near Real-time Spraying Progress Monitoring
Primary Actor	HTX Manager
Secondary Actor	System (auto-update)
Preconditions	1. At least 1 spraying order is in 'In progress' status (checked in). 2. The HTX Manager is logged into the Web Dashboard.
Postconditions (Success)	The HTX Manager has full near real-time information on the progress of the entire drone fleet.
Main Flow	1. The HTX Manager opens the 'Monitoring Map' Dashboard. 2. The system displays the positions of all flying drones on the map (updated ≤ 3 seconds). 3. The HTX Manager selects a drone to view details: % area completed, remaining pesticide, estimated completion time. 4. The HTX Manager can view the drone's flight trail. 5. The system displays a red alert if a drone approaches or crosses the geofence boundary.
Alternative Flow	Alt 2a - Drone connection lost (> 15 seconds without signal): The system displays a 'Disconnected' status and alerts the HTX Manager. Alt 5a - Drone leaves the geofence: The system immediately triggers an audio alert + popup on the dashboard and sends a push notification to the Pilot.
Business Rule	Position data is logged with a timestamp every 3 seconds to support later legal reporting.
Priority	Must - Linked: FR10, FR11, FR12, FR13

UC05 - Receive and Handle Low-Pesticide / Low-Battery Alerts

Use Case ID	UC05
Use Case Name	Receive and Handle Low-Pesticide / Low-Battery Alerts
Primary Actor	Operator / Pilot
Secondary Actor	HTX Manager
Preconditions	1. The Pilot is performing spraying (checked in UC03). 2. Alert thresholds are configured (battery: 20%, pesticide: enough to finish the current zone).
Postconditions (Success)	The alert is logged with a timestamp. Order status is updated accordingly (Paused / In progress / Completed early).
Main Flow	1. The system detects (from operator manual entry on the Mobile App or from structured simulated telemetry, per Section 2.5 / FR11) drone batteries dropping below 20% (or pesticide insufficient for the next zone). 2. The system sends a push notification to the Pilot and an alert on the HTX Manager's dashboard. 3. The Pilot confirms receipt of the alert. 4. The Pilot lands, refills pesticide / replaces battery, then taps 'Resume spraying'.

	5. The system records the pause time and recalculates the completion percentage.
Alternative Flow	Alt 3a - The Pilot does not respond within 5 minutes: The system escalates the alert to the HTX Manager and turns the drone icon red. Alt 4a The Pilot decides to terminate spraying entirely: The Pilot taps 'End early', and the system records the reason and the area already sprayed.
Business Rule	Battery and pesticide alert thresholds are configured by the HTX Manager; defaults: battery 20%, pesticide 15%.
Priority	Must Linked: FR10, FR11

UC06 - Adverse Weather Alert and Handling of In-flight Orders

Use Case ID	UC06
Use Case Name	Adverse Weather Alert and Handling of In-flight Orders
Primary Actor	System (automatic)
Secondary Actor	HTX Manager, Pilot
Preconditions	1. At least 1 spraying order is in progress or scheduled to start (within the next 2 hours). 2. The system is integrated with weather data (API).
Postconditions (Success)	Decisions and their rationale are fully logged. The Pilot receives the final decision notification.
Postconditions (Failure)	If the Weather API does not respond for more than 30 minutes (see Dependency in Section 2.5), the system switches to a 'Weather alert - Unavailable' state and displays a banner for the HTX Manager to make a manual decision; orders are NOT automatically suspended. If neither the HTX Manager nor the Pilot responds to an alert within 30 minutes when wind ≥ 3 m/s, the system forces the order status to 'Suspended - Lacking safety confirmation' to safeguard legal compliance.
Main Flow	1. The system pulls weather data every 15 minutes from the API. 2. The system detects wind speed ≥3 m/s or rain forecast within the next 30 minutes. 3. The system automatically sends alerts to the Pilot (in flight) and the HTX Manager. 4. The HTX Manager reviews and decides: halt spraying or continue with risk acknowledgement. 5. If halted: the system updates the order status to 'Suspended - Weather' and suggests an alternative time slot.
Alternative Flow	Alt 4a - The HTX Manager does not respond within 10 minutes: The system automatically marks the order as 'Needs review' and continues to send reminders every 5 minutes. Alt 4b - The HTX Manager confirms continuation: The system records the decision + timestamp + confirmer's name in the audit trail.
Business Rule	Wind threshold: ≥ 3 m/s (an assumed threshold based on DJI Agras agricultural-drone spraying guidance, pending expert confirmation before field use). This threshold is a SOFT constraint system-wide: the HTX Manager may override it with a risk acknowledgement logged in the audit trail (consistent with UC01); if no one responds within 30 minutes, the order is force-suspended (see Postconditions). Weather data from the OpenWeatherMap API or equivalent.
Priority	Should - Linked: FR11, FR17

UC07 - Check-out and Spraying Completion Record

Use Case ID	UC07
Use Case Name	Check-out and Spraying Completion Record
Primary Actor	Operator / Pilot
Secondary Actor	System
Preconditions	1. The Pilot has checked in (UC03) and finished spraying on-site. 2. The Pilot has network connectivity (or is offline - will sync later).
Postconditions (Success)	A complete flight log is saved to the database with full legal information. The Farmer receives a completion notification. Pesticide inventory is updated.
Main Flow	1. The Pilot taps 'Complete spraying' on the Mobile App. 2. The Pilot enters actual data: pesticide used, actual area sprayed, abnormal observations (if any). 3. The system calculates and displays a summary: sprayed area, pesticide used, flight duration, comparison with the plan. 4. The Pilot confirms via the authenticated session (RBAC per FR18); the confirmation is recorded in the audit log with timestamp and Pilot ID. 5. The system automatically generates the flight log and updates the order status to 'Completed'. 6. The system sends a notification to the Farmer via SMS/Zalo.
Alternative Flow	Alt 2a - Actual pesticide usage deviates > 20% from the plan: The system requires the Pilot to enter a reason before confirmation. Alt 4a - Offline: The system stores the check-out locally and syncs within ≤ 30 seconds once connectivity returns.
Business Rule	The flight log includes drone identifier, pilot ID, GPS trail, pesticide volume - mandatory under Decree 288/2025.
Priority	Must - Linked: FR09, FR13, FR16, FR17

UC08 - Export Plant Protection Department Compliance Report

Use Case ID	UC08
Use Case Name	Export Plant Protection Department Compliance Report
Primary Actor	HTX Manager
Secondary Actor	Plant Protection Department (report recipient)
Preconditions	1. The HTX Manager is logged in. 2. Spraying log data exists for the reporting period (monthly/quarterly/seasonal).
Postconditions (Success)	The PDF/Excel report has been downloaded. The export history is saved to the system's audit log.
Main Flow	1. The HTX Manager selects 'Export compliance report' from the Compliance menu. 2. The HTX Manager selects the time range and target cooperative.

	3. The system aggregates data: list of drones (with identifiers), list of pilots (with certificate numbers), total spray area per pesticide type. 4. The system generates the PDF and Excel report files simultaneously. 5. The HTX Manager previews and downloads the report. The system saves export history (who, when).
Alternative Flow	Alt 3a - A pilot flew but their certificate expired during the period: The system highlights the violating flights in red in the report and creates a 'Compliance note' section. Alt 4a - Insufficient data (missing logs): The system lists the spraying orders with missing data and refuses to export the report until they are complete.
Business Rule	The report layout is aligned with the Plant Protection Department template (FR-REG02); the exact official format is subject to authority confirmation. The report must be digitally signed or confirmed by the HTX Manager.
Priority	Must - Linked: FR13, FR14, FR15

UC09 - Field Map Management (Field Mapping)

Use Case ID	**UC09**
Use Case Name	Field Map Management (Field Mapping)
Primary Actor	HTX Manager / Agronomist
Secondary Actor	---
Preconditions	1. The user is logged in with the role of HTX Manager or Agronomist. 2. The Farmer is registered in the system.
Postconditions (Success)	The field is saved with complete coordinates, area, and spray formula. The Farmer can book spraying for this field.
Postconditions (Failure)	If the polygon is not closed or the area is outside the limits (< 0.1 ha or > 100 ha), the system does not save the field and displays a specific error; the Farmer's account is not associated with the erroneous field. If an overlap with an existing field is detected (Alt 3a) and the user does not accept an adjustment, the operation is cancelled and the map is rolled back to its previous state (safe rollback).
Main Flow	1. The HTX Manager selects 'Add new field'. 2. The user draws the field boundary on the map (or enters GPS coordinates). 3. The system automatically computes the area (ha) from the coordinates. 4. The user enters information: field name, owner (Farmer), crop type, seasonal schedule. 5. The Agronomist configures the spray formula: pesticide type, dosage (L/ha), number of sprays per season. 6. The system saves the map and associates it with the Farmer's account.
Alternative Flow	Alt 2a - Invalid manual GPS coordinates: The system reports a format error and requests re-entry. Alt 3a - The new field overlaps an existing one: The system warns and displays the overlapping region on the map.
Business Rule	Field boundaries must be closed polygons. Minimum area: 0.1 ha. Maximum: 100 ha per field.
Priority	Must - Linked: FR01, FR02

UC10 - View Spraying History and Export VietGAP Data

Use Case ID	UC10
Use Case Name	View Spraying History and Export VietGAP Data
Primary Actor	Farmer / HTX Manager (HTX Manager exports on behalf of a contracted Export Company; Export Company is not a system actor in the MVP RBAC - see Section 3 D1, exporter integration planned post-MVP)
Secondary Actor	---
Preconditions	1. The user is logged in. 2. The field has at least 1 complete spraying record.
Postconditions (Success)	Spraying history data is displayed/exported successfully. Access is logged.
Main Flow	1. The user selects the field whose history they want to view. 2. The system displays spraying sessions: date, pesticide type, dosage, pilot, drone, area. 3. The user filters by time range or pesticide type. 4. The HTX Manager (on behalf of the contracted export company) selects 'Export VietGAP/GlobalGAP data'. 5. The system generates an Excel/PDF file with full traceability information.
Alternative Flow	Alt 4a - The contracted export company has not yet been granted a data-sharing scope: the HTX Manager initiates an internal access-approval workflow (recorded in the audit log) before exporting and sharing data offline.
Business Rule	Data is only shared after HTX Manager approval. Exports are recorded in the audit log: who, when, what data.
Priority	Should - Linked: FR14, FR15

5.1 Traceability Matrix FR, UC

The table below is the reverse of the 'Linked FR' column shown in each UC: each FR is traced to the UC(s) that use it (Direct UC coverage), or otherwise classified as Cross-cutting (UI/i18n/RBAC/Test) where it does not bind to a single UC. This table supports consistency checking and is a ready reference for the Oral Vivas.

FR ID	Short Description	Priority	UCs Using This FR
FR01	Create/edit field maps (Field Mapping)	**Must**	UC09
FR02	Calculate per-zone pesticide dosage	**Must**	UC09
FR03	Book spraying in ≤ 5 steps (Farmer)	**Must**	UC01
FR04	Schedule-conflict alerts	**Must**	UC01
FR05	Calendar view (HTX Manager)	**Should**	UC01, UC02 (indirectly)
FR06	Rule-based drone-assignment suggestions	**Must**	UC02
FR07	Pilot records (certificate, expiration)	**Must**	UC02, UC08
FR08	Assignment notifications to the Operator	**Must**	UC02
FR09	GPS-verified Check-in/Check-out	**Must**	UC03, UC07
FR10	Near real-time map ≤ 3s	**Must**	UC03, UC04, UC05

FR ID	Short Description	Priority	UCs Using This FR
FR11	Battery / pesticide / geofence / weather alerts	**Must**	UC04, UC05, UC06
FR12	Geofencing (50 m buffer around power lines)	**Must**	UC03, UC04
FR13	Complete flight logs (Decree 288)	**Must**	UC02, UC04, UC07, UC08
FR14	Export Plant Protection Department reports	**Must**	UC08, UC10
FR15	Export VietGAP/GlobalGAP	**Should**	UC08, UC10
FR16	Manage pesticide inventory	**Should**	UC07
FR17	SMS/Zalo notifications to Farmers	**Should**	UC01, UC06, UC07
FR18	RBAC permissions (5 roles)	**Must**	All UCs (cross-cutting authorization)
FR19	Dark mode	**Could**	Cross-cutting UX (not tied to a specific UC)
FR20	Multi-language VI/EN	**Could**	Cross-cutting i18n (not tied to a specific UC)
FR21	Drone master records (ID, model, tank, status)	**Must**	UC02 (drone status read for assignment), UC08 (drone identifier list in compliance report)

Note: Coverage Type classification: 18 FRs have Direct UC coverage (FR01-FR04, FR06-FR17, FR21); FR05 (calendar view) has Indirect coverage via UC01/UC02; FR18 (RBAC) is Cross-cutting authorization, used implicitly in every UC via the ‘is logged in’ precondition; FR19 (Dark Mode) and FR20 (Multi-language VI/EN) are Cross-cutting UI/i18n concerns with priority Could, not tied to a single UC; FR21 (Drone Registry) is a prerequisite data-management function read by UC02 and UC08. Cross-cutting FRs will be verified through UI mockups in D4 and test cases in D4 QA Plan rather than a single UC scenario.

6. User Stories

The User Stories below are written following the **INVEST** standard with acceptance criteria in the **Given-When-Then** format. Priority follows MoSCoW; Size is relatively estimated (S/M/L).

Story ID	US01
User Story	As a Farmer, I want to book a spraying session in ≤ 5 taps so that I can schedule without calling the HTX office.
Acceptance (Given)	The Farmer is logged into the Mobile App and has a field created in the system.
Acceptance (When)	The Farmer selects a field → selects date/time → selects a pesticide → taps 'Confirm'.
Acceptance (Then)	The system displays a booking confirmation and sends an SMS confirmation to the Farmer within <30 seconds.
Priority / Size	Must | S (Small)
Linked FR	FR03, FR04, FR17

Story ID	US02
User Story	As an HTX Manager, I want the system to auto-suggest the best drone and pilot for each job so that I don't have to manually match resources.
Acceptance (Given)	There is a spraying order pending assignment, and ≥ 1 drone and pilot are available with valid certificates.
Acceptance (When)	The HTX Manager opens the assignment screen and views the pending order.
Acceptance (Then)	The system suggests the most suitable drone and pilot within < 3 seconds. The HTX Manager only needs to tap 'Confirm' or override the suggestion.
Priority / Size	Must | M (Medium)
Linked FR	FR06, FR07, FR08

Story ID	US03
User Story	As an Operator/Pilot, I want to check-in on-site with GPS verification so that my flight log is automatically recorded without paperwork.
Acceptance (Given)	The Pilot is near the field (within 100m) and has a flight scheduled today in the Mobile App.
Acceptance (When)	The Pilot taps 'On-site Check-in' in the Mobile App.
Acceptance (Then)	The system verifies GPS within < 5 seconds, records the check-in with timestamp and location, and updates the order status to 'In progress'.
Priority / Size	Must | S (Small)
Linked FR	FR09, FR10, FR13

Story ID	US04
User Story	As an HTX Manager, I want to see all drones on a near real-time map updated every 3 seconds so that I can monitor the fleet from the office.
Acceptance (Given)	There is ≥ 1 drone in 'In progress' status currently flying.
Acceptance (When)	The HTX Manager opens the 'Monitoring Map' Dashboard.
Acceptance (Then)	All drones are shown on the map at their accurate positions, updated ≤ 3 seconds. Clicking a drone reveals % completion and remaining pesticide.
Priority / Size	Must | M (Medium)
Linked FR	FR10, FR11, FR12

Story ID	US05
User Story	As an HTX Manager, I want to export a compliance report for the Plant Protection Department in PDF/Excel format so that I can meet the reporting deadline under Decree 288/2025.
Acceptance (Given)	The HTX Manager is logged in and has complete spraying-log data for the reporting period.
Acceptance (When)	The HTX Manager selects the reporting period and taps 'Export compliance report'.
Acceptance (Then)	The system generates PDF and Excel files within < 30 seconds containing complete information: drone identifier, pilot certificate number, sprayed area, pesticide type. The file layout is aligned with the Plant Protection Department's reporting fields (exact official format subject to authority confirmation).
Priority / Size	Must | L (Large)
Linked FR	FR13, FR14

Story ID	US06
User Story	As an Operator, I want the Mobile App to work offline so that I can check-in and check-out even when there's no 4G signal at the field.
Acceptance (Given)	The Pilot is in an area without 4G but has a pre-synced flight schedule.
Acceptance (When)	The Pilot performs check-in or check-out while offline.
Acceptance (Then)	The app saves the action locally and displays 'Will sync when online'. Once 4G is restored, it syncs automatically within ≤ 30 seconds.
Priority / Size	Must | M (Medium)
Linked FR	FR09, NFR06

Story ID	US07
User Story	As an Agronomist, I want to configure spray formulas per field zone so that the correct pesticide dosage is automatically calculated for each spraying job.
Acceptance (Given)	The Agronomist is logged in, and the field already has a map.
Acceptance (When)	The Agronomist selects a field and opens the 'Configure spray formula' tab.
Acceptance (Then)	The Agronomist enters pesticide type, dosage (L/ha), and number of sprays. The system automatically computes the total pesticide needed for the whole field and shows a preview.
Priority / Size	Should | M (Medium)
Linked FR	FR01, FR02

Story ID	US08
User Story	As a Farmer, I want to receive SMS/Zalo updates when my field has been sprayed so that I don't have to call to confirm.
Acceptance (Given)	The Farmer's spraying order has been completed (UC07).
Acceptance (When)	The Pilot checks out and confirms completion.
Acceptance (Then)	The Farmer receives an SMS/Zalo message within < 1 minute with the content: 'Field [name] has been sprayed at [time]. Pesticide used: [X] liters.'
Priority / Size	Should | S (Small)
Linked FR	FR17

Appendix A - Glossary (UAV Domain Terminology)

Term / Abbreviation	Definition
AgriUAV Platform	System name - a web and mobile platform that coordinates precision agricultural drone spraying in the Mekong Delta.
Mekong Delta (ĐBSCL)	Đồng bằng sông Cửu Long - Vietnam's largest specialized rice-growing region.
Drone / UAV	Unmanned Aerial Vehicle — in this document refers to agricultural spraying drones (DJI Agras, XAG, etc.).
HTX (Cooperative)	Agricultural cooperative - an organization that manages and provides drone spraying services to farmers.
Operator / Pilot	The person who operates the drone directly in the field, holding a valid flight certificate under Decree 288/2025.
Geofencing	A virtual geographic fence - a boundary defining the safe flight area, triggering an alert when a drone crosses it.
Field Mapping	Digitization of field maps with GPS coordinates and accurate area.
Check-in / Check-out	The Pilot confirms on-site presence (check-in) and confirms spraying completion (check-out) via the GPS-verified Mobile App.
Decree 288/2025/ND-CP	Decree 288/2025/ND-CP - regulating the operation of unmanned aerial vehicles, effective from 05/11/2025.
Plant Protection Department	Plant Protection Department - the state agency in charge of pesticides and crop-pest control.
Operations Department (General Staff)	A unit under the General Staff that issues flight permits and manages drone identifiers under Decree 288/2025.
MAVLink	Micro Air Vehicle Link - a common drone communication protocol, used to receive telemetry (position, battery, speed).
RTK GPS	Real-Time Kinematic GPS - high-accuracy GPS (≤ 2cm), commonly integrated into high-end agricultural drones.
VietGAP / GlobalGAP	Vietnamese / international Good Agricultural Practice standards - require traceability of pesticide usage.
SaaS	Software-as-a-Service is a software subscription business model billed monthly/yearly (500,000 VND/drone/month).
RBAC	Role-Based Access Control - permissions assigned by role: Admin, HTX Manager, Operator, Farmer, Agronomist.
STRIDE	Security analysis model: Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege.
TLS 1.3	Transport Layer Security v1.3 - the latest standard internet connection encryption protocol.
BVLOS	Beyond Visual Line of Sight - flight beyond direct line of sight (outside the scope of this system).
SOM / SAM / TAM	Serviceable Obtainable / Addressable Market / Total Addressable Market - market-sizing segmentation tiers.

7. AI Artifact Audit Log | Entries #09-13 (W4 / Phase 2)

This section documents AI interactions during Phase 2 (Requirements). Entries continue the cumulative log from D1 (entries #01-08), targeting ≥ 25 entries total by end of W10.

No.	Week / Phase	DTC Component	AI Tool	Prompt (Full Context)	AI Output (Summary)	Decision & Human Delta	Verification
09	W4 / P2 Requirements	Process / Algorithm	Claude	Design a requirement elicitation workflow for an Agricultural UAV project: steps from stakeholder identification to SRS finalization, including techniques for each step (interview, observation, prototyping...).	Proposed a 5-step workflow: Identify stakeholders → Elicit requirements → Analyze → Specify → Validate. Techniques: structured interview, JAD session, use case workshop.	Accepted the workflow structure. Added 'Domain immersion' step: read Cuc BVTV reports and Nguyen 2025 BEFORE interviews. Added 'Roleplay interview': team acts as farmer/HTX manager since direct stakeholder access is limited. Note: AI did not suggest role play for students with no field access.	Role-play interview confirmed in W4 instructor guidance. Domain immersion cross checked with NFR Availability.
10	W4 / P2 Requirements	Decomposition	Claude	Propose Functional Requirements for AgriUAV Platform, modules: Field Mapping, Spray Scheduling, Fleet Dispatch, Real-time Monitoring. Each FR has ID, description, priority (Must/Should/Could).	Listed 18 FRs classified by 4 modules. Covered GPS mapping, schedule creation, drone assignment, telemetry display.	Accepted 14 core FRs. Removed 2 FRs: 'AI pest detection' and 'autonomous flight path optimization' out of IS scope (AI/robotics). Merged 2 FRs about 'drone status display' into FR10. Added 2 regulatory FRs AI missed: FR13: Store drone ID + pilot cert per Decree 288 (FR-REG01) FR14: Export reports in Plant Protection Dept. format (FR-REG02)	FR-REG01/02 verified directly against Decree 288/2025/ND-CP text and Cuc Trong trot & BVTV requirements.
11	W4 / P2 Requirements	Process / Algorithm	Claude	Write Non-Functional Requirements for Agricultural UAV Platform with quantitative thresholds: Performance (drone	Suggested NFRs: update ≤ 5 seconds, uptime 95%, usability 10 min/task, 100	Accepted NFR structure calibrated for DBSCL context: Performance: ≤5s -> ≤3s	3s threshold based on DJI Agras T40 actual speed. Age ref: Nguyen

				position update), Reliability (uptime during season), Usability (limited IT farmers), Scalability, Security.	drones concurrent, AES-256 encryption.	(drone moves ~5-7 m/s; 5s too slow for safety alerts) • Reliability: 95% -> 99% (peak season 30-60 days; downtime unacceptable) • Usability: 10 min → ≤ 5 min (Nguyen 2025: avg age ~51, limited IT) • Scalability: kept the 50-drone reference workload per HTX (AI suggested 100, beyond the evaluation baseline) Added NFR06 Availability: offline mode when no 4G AI omitted entirely	2025 N=940. Offline req cross-checked with Mekong Delta field conditions.
12	W4 / P2 Requirements	Decomposition	Claude	List ≥ 8 Use Cases for AgriUAV Platform with actor, precondition, main flow (3-5 steps), alternative flow, postcondition. Focus on spray scheduling, drone dispatch, real time monitoring, alerts.	Generated 10 use cases with full main flow but only 2/10 UCs had alternative flows. UC about regulatory reporting was missing entirely.	Accepted 9 UCs. Removed UC from 'User Registration' too simple, not a standalone UC. Added missing alt flows for 6 UCs: • UC03: Alt flows for GPS outside 100m radius and offline check-in (no 4G) • UC04: Alt flow when 4G connection lost mid-session • UC05: Alt flow when operator does not respond in 5 min • UC06: Alt flow when weather changes while drone is in-flight (manager override + audit trail) • UC07: Alt flow when actual pesticide usage deviates > 20% from plan (reason required)	Alt flow 'offline 4G' cross-checked with NFR06. UC10 from FR15 (VietGAP) and Decree 288/2025 audit trail requirements.

						• UC08: Alt flow when a pilot certificate expired during the reporting period (compliance note) Added UC10: View Spraying History & Export VietGAP AI missed regulatory traceability UC entirely	
13	W4 / P2 Requirements	DTC component - Pattern Recognition / Consistency & Counter-argument	Claude	Review D2 v2.4.3 against the D1 v1.5.1 proposal and the RBL workshop criteria; identify scope/actor/telemetry/traceability/validation inconsistencies and propose concrete corrections.	Identified 15 issues across 4 severity levels: telemetry semantics mismatch between Scope (mobile-GPS proxy / simulated) and FR10-FR12 / UC04-UC06 (described drone position, battery, pesticide as real sensor data); UC10 used “Export Company” as a system actor outside the 5-role RBAC; UC03 Alt 3a allowed Pilot self-bypass of the 100m GPS check; UC05 battery/pesticide data source undefined; traceability statement claimed “no orphaned FR” despite cross-cutting FR19/20; NFR05 mixed requirement with technology choice; NFR04 mixed 1-HTX baseline with	Accepted the diagnostic; applied surgical corrections rather than rewriting. (a) Telemetry: clarified data source in FR10, FR11 and UC05 (mobile-GPS proxy or structured simulated telemetry within MVP); MAVLink remains post-MVP per Section 2.5. (b) UC10 actor: changed Primary Actor to HTX Manager acting on behalf of the contracted Export Company, consistent with the 5-role RBAC (FR18) and D1’s post-MVP exporter integration. (c) UC03 Alt 3a: removed Pilot self-bypass; HTX Manager approval required with audit-trail entry. (d) UC07: replaced fingerprint/PIN with authenticated-session confirmation already covered by FR18, eliminating the hidden	Each correction was cross-checked against D1 v1.5.1 (canonical) and the RBL workshop checklist (Part 2: RQ → method → evaluation; Part 3: verifiability). Counts confirmed after edits: 21 FR, 7 NFR, 10 UC, 8 US, 5 RBAC roles - all preserved. Reflection note: this pass illustrates AI-assisted consistency auditing where the human reviewer (D2 evaluator report) caught artefacts the original AI elicitation passes did not surface; the Human Delta is the choice between option (A) HTX Manager export vs

					10-HTX planning; legal retention and report format lacked specific article references; UC07 introduced biometric/PIN as a hidden requirement; FR06 rule set narrower than UC02.	requirement. (e) FR06: added battery, pilot certificate validity, and pilot schedule to the rule set (source-of-truth alignment with UC02). (f) NFR04: separated MVP acceptance baseline (1 HTX, ~20 req/s) from a future-cluster planning assumption (10 HTX, ~200 req/s). (g) NFR05: rephrased as solution-neutral security outcomes with current minimum baselines, deferring final technology choice to D4. (h) Retention 5y and report format relabelled as working assumptions pending authority confirmation. (i) Traceability: dropped “no orphaned” phrasing; introduced explicit Coverage Type classification (Direct / Indirect / Cross-cutting). (j) Added Section 2.2 fix removing “automatic confirmation” (inconsistent with UC01 Pending-confirmation flow). (k) Section 1.1 “signed agreement” softened to “baseline working agreement, pending	option (B) adding External Partner role - option (A) was chosen as more scope-conservative.

						stakeholder sign-off”.	

Appendix B - Requirements Research & Validation Summary (RBL)

This short appendix records the Research-Based Learning (RBL) framing for D2 without altering the IEEE 830 body of the SRS. It documents the Phase-Requirements research questions, objectives, validation methods, and evaluation criteria used to produce and check the requirements set.

B.1 Research Questions (Phase: Requirements)

RQ-D2.1 Which Functional and Non-Functional Requirements are necessary to address the operational, safety, and compliance pain points identified in D1 within the constraints of an MVP that does not control drone hardware?

RQ-D2.2 How do Mekong-Delta-specific constraints (intermittent 4G connectivity, low-IT-skill end-users, Decree 288/2025/ND-CP, telemetry-without-MAVLink) shape the requirements set?

RQ-D2.3 Is the resulting requirements set complete, consistent, verifiable, and traceable enough to serve as the baseline for D3 System Modelling and D4 System Design?

B.2 Phase Objectives

(O1) Elicit and classify requirements from D1 stakeholders, pain points, regulations, and ĐBSCL constraints. (O2) Translate them into verifiable FR, NFR, Use Cases, and User Stories with explicit IDs, priority, and source. (O3) Cross-check consistency across Scope, FR, NFR, UC, User Stories, and Interfaces (including telemetry source and actor model). (O4) Establish a traceability baseline that downstream D3 models and D4 design can rely on, classified by coverage type. (O5) Record AI-assisted decisions and Human Delta in the Audit Log, including at least one consistency-correction entry for the phase (Entry #13 above).

B.3 Validation Methods Used

(M1) Document analysis - D1 v1.5.1, Decree 288/2025/ND-CP, Nguyen et al. (2025), Tuoi Tre (2024) incident report - mapped to requirement IDs in the Source column of Sections 3.1 and 3.2. (M2) Role-play stakeholder interviews - team members assumed Farmer / HTX Manager / Operator / Agronomist personas under instructor guidance in W4; question lists, raised conflicts, and accepted changes are kept as internal study notes (sign-off pending future real-cooperative pilot, recorded as a study limitation in D1 §7 Risk R2). (M3) Scenario walkthrough of UC01, UC02, UC03, UC04, UC06, and UC08 to surface alternative flows and failure post-conditions. (M4) Consistency audit across Scope ↔ FR ↔ NFR ↔ UC ↔ Interface, performed in this v2.4 pass and documented in Audit Log entry #13. (M5) Traceability review including reverse matrix and Coverage Type classification (Section 5.1). (M6) AI Reflection audit - cumulative entries #09-#13 with prompt, output, decision, and verification columns.

B.4 Evaluation Criteria for Requirement Quality

Each requirement was assessed against seven criteria: Correctness (reflects a real stakeholder need or domain rule), Completeness (no missing actor, data source, exception or constraint), Consistency (no contradiction with Scope / other FR / NFR / UC / Interface), Unambiguity (no undefined terms such as “real-time” or “secure” without a quantitative threshold or definition), Verifiability (testable through a threshold, scenario, or Given-When-Then acceptance criterion), Traceability (linked to a source upstream and an artefact downstream), and Feasibility (achievable within the MVP scope, which excludes direct drone-hardware integration). The v2.4 correction pass focused primarily on Consistency, Verifiability, and Traceability gaps identified by the D2 evaluator report.

B.5 D2 Contribution Summary

D2 contributes: (i) a complete IEEE 830-aligned requirement baseline (21 FR, 7 NFR, 10 UC, 8 User Stories) consistent with the D1 problem framing; (ii) an FR/UC traceability matrix with explicit Coverage Type classification (Direct / Indirect / Cross-cutting); (iii) evidence-bounded compliance and telemetry language that distinguishes verified facts, working assumptions, and post-MVP scope; (iv) a cumulative AI Audit Log (entries #09-#13) including a consistency-correction entry; and (v) this RBL appendix recording the research questions, objectives, methods, and evaluation criteria so that D3 modelling and D4 design can begin from an auditable baseline.